

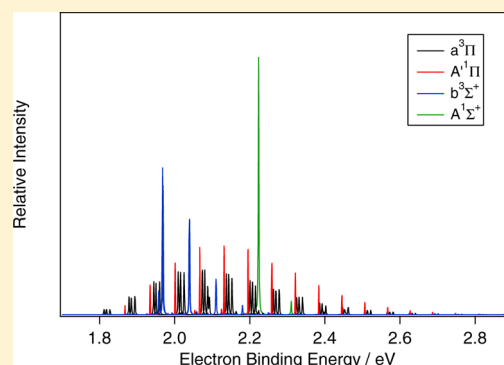
Theoretical Investigations on CaO Ions: Vibronic States and Photoelectron Spectroscopy

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ABSTRACT: The low-lying electronic states, $X^2\Pi$ and $A^2\Sigma^+$ of CaO^+ and $X^2\Sigma^+$ and $A^2\Pi$ of CaO^- , have been determined at the MRCI+Q level of theory with the aug-cc-pV5Z(O) and cc-pCV5Z(Ca) basis sets. The two states of CaO^+ are close within <0.1 eV and coupled via spin–orbit effect. The $X^2\Sigma^+$ and $A^2\Pi$ states of CaO^- are energetically separated by <1 eV such that the first excited state is close to the electronic ground state of neutral CaO and unstable with respect to electron detachment. Using the potential energy curves and the spin–orbit coupling terms, the vibronic energy levels of these ions have been determined. The ionization energy and the electron affinity of CaO are calculated at 6.79 and 0.79 eV, respectively. The photoelectron spectra of CaO^- and CaO have also been simulated.



1. INTRODUCTION

Neutral calcium monoxide (CaO) has been widely studied in the gas phase. This diatomic exhibits a $X^1\Sigma^+$ electronic ground state with a strong multiconfigurational character in the equilibrium region due to its dominant ionic nature.¹ Recently, electronic structure calculations have been performed on the neutral CaO, resulting in accurate potential energy curves for its low-lying electronic states.¹ These curves were then used in calculations including spin–orbit (SO), rotational, and vibronic couplings for the determination of the rovibronic states of this diatomic up to about 15 000 cm^{-1} above the ground electronic minimum.²

In contrast with the neutral CaO, relatively few studies are related to the CaO^+ and CaO^- species. From a theoretical point of view, the most reliable works are those reported by Partridge et al.³ and by Harrison et al.⁴ In the former one,³ the two low-lying electronic states of CaO^+ , namely, the $X^2\Pi$ and $A^2\Sigma^+$ states, have been investigated using single and double configuration interaction (CISD). These two states have been found to be separated by a few millielectronvolts, and their main spectroscopic characteristics (i.e., equilibrium geometries and harmonic wavenumbers) were reported. In the most recent work of Harrison et al.,⁴ the low-lying electronic states of CaO, CaO^+ , and CaO^- were investigated using restricted coupled-cluster approach (RCCSD(T)). For the cation, the energy difference between the X and A states was confirmed and the main spectroscopic constants were calculated. For both the X and A states, the harmonic wavenumber differs by $\sim 30 \text{ cm}^{-1}$ from results of Partridge et al.,³ but the equilibrium geometries are in agreement. The ionization energy (IE) of CaO was reported by Harrison et al.⁴ at 6.767 eV, in agreement with the values ranging from 6.5 to 6.7 eV deduced from mass spectrometric studies.^{5–7}

However, neither Partridge et al.³ nor Harrison et al.⁴ considered the SO couplings between these X and A states of CaO^+ . Such couplings are nevertheless mandatory for an accurate determination of the vibronic states of such systems,² in particular, when the electronic states have their minima close in energy.

For the CaO^- anion, the only information available in the literature are the coupled-cluster computations of Harrison et al.⁴ Two electronic states, namely, $X^2\Sigma^+$ and $A^2\Pi$, were investigated by these authors. The equilibrium geometries and the harmonic wavenumbers of both states were reported, and the electron affinity (EA) of CaO was calculated at 0.865 eV. The energy difference between the X and A states was also computed at 0.72 eV. To the best of our knowledge, this anion has not yet been observed. It could, however, be involved during the fragmentation of the experimentally characterized Ca_2O triatomic molecule,^{8,9} which was also recently studied theoretically.¹⁰ A predominant $\text{Ca}^+\text{O}^{2-}\text{Ca}^+$ ionic character was noticed, and a diabatic dissociation into $\text{CaO}^- + \text{Ca}^+$ is expected.

The aims of the present work are to supplement our previous studies on neutral CaO by investigating the ions. Their lower electronic states are calculated at the same level of theory, and the corresponding vibronic energy levels are determined by including SO couplings when required. To be predictive, photoelectron spectra are also simulated from the electronic ground states of CaO and CaO^- . The manuscript is organized as follows: the next section describes the ab initio calculations of the lowest electronic states of CaO^+ and CaO^- , then Section 3 is

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devoted to the determination of the vibronic states of both ions. Finally, photoelectron spectra of CaO^- and CaO are discussed in Section 4.

2. LOW-LYING ELECTRONIC STATES OF CaO^+ AND CaO^-

In the first part of this study, the low-lying electronic states of CaO^+ converging to the two lowest dissociation limits and the two lowest electronic states of CaO^- have been investigated at the complete active space self-consistent field (CASSCF) level of theory.^{11,12} We used the same active space as in our studies of the neutral molecule, that is, the $(11\sigma, 5\pi, 1\delta)$ molecular orbitals (MOs) were considered, with the $(4\sigma, 1\pi)$ MOs frozen and the $(7\sigma, 2\pi)$ MOs restricted to be doubly occupied. The aug-cc-pV5Z basis set¹³ and the cc-pCV5Z basis set¹⁴ were chosen for the description of the oxygen and calcium atoms, respectively. All of these computations were performed in the C_{2v} symmetry group with the MOLPRO program.¹⁵ This methodology is identical to the one employed for the neutral system and is described in detail in ref 1.

Figure 1 presents the resulting electronic states of CaO^+ and CaO^- , together with the electronic states of the neutral CaO

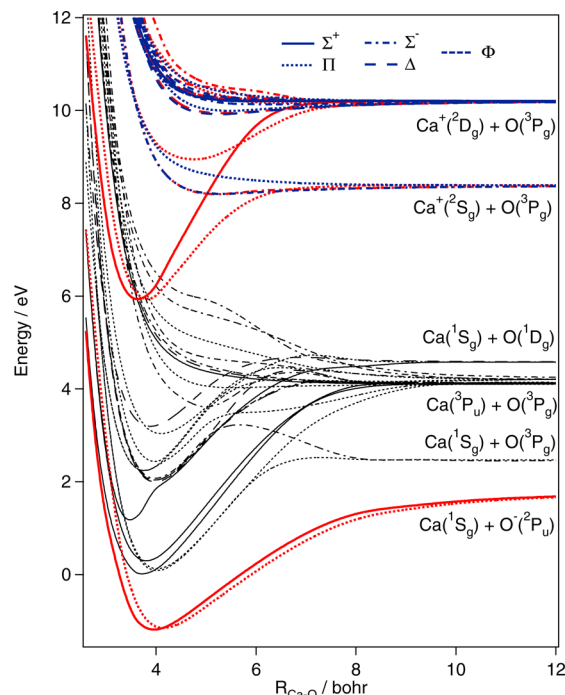


Figure 1. CASSCF potential-energy curves of the low-lying electronic states of CaO (in black), CaO^+ (doublets are in red and quartets are in blue), and CaO^- (doublets in red).

taken from ref 1 and computed at the same level of theory. The nature of the corresponding electronic states is listed in Table 1. As in the neutral molecule, the two lowest electronic states of both ions are found to have similar equilibrium energies and bond lengths at the CASSCF level. Indeed for CaO^- , the X and A states are separated by <0.1 eV, whereas these states are quasi-degenerate for CaO^+ . At this level of calculations, the X and A states of CaO^- appear to be well-isolated from the low-lying electronic states of the neutral molecule. For CaO^+ , most of the electronic states are repulsive (or have a shallow minimum), except for the X, A, and the second $^2\Pi$ states, which are bound.

Table 1. Nature of the Low-Lying Electronic States of CaO , CaO^+ , and CaO^- , Converging to Their Lowest Dissociation Limits

molecule	asymptote	molecular state
CaO^-	$\text{Ca}(^1S_g) + \text{O}^-(^2P_u)$	$^2\Sigma^+, ^2\Pi$
CaO	$\text{Ca}(^1S_g) + \text{O}(^3P_g)$	$^3\Sigma^-, ^3\Pi$
	$\text{Ca}(^3P_u) + \text{O}(^3P_g)$	$1,3,5\Sigma^+, 1,3,5\Sigma^-(2), 1,3,5\Pi(2), 1,3,5\Delta$
	$\text{Ca}(^1S_g) + \text{O}(^1D_g)$	$1\Sigma^+, 1\Pi, 1\Delta$
CaO^+	$\text{Ca}^+(^2S_g) + \text{O}(^3P_g)$	$2,4\Sigma^-, 2,4\Pi$
	$\text{Ca}^+(^2D_g) + \text{O}(^3P_g)$	$2,4\Sigma^+, 2,4\Sigma^-(2), 2,4\Pi(3), 2,4\Delta(2), 2,4\Phi$

The $A^2\Sigma^+$ state dissociates to the second dissociation limit of CaO^+ (i.e., $\text{Ca}^+(^2D_g) + \text{O}(^3P_g)$) and is crossed by three repulsive states, which are correlating to the first dissociation limit (i.e., $\text{Ca}^+(^2S_g) + \text{O}(^3P_g)$). These states are of $^4\Pi$, $^4\Sigma^-$, and $^2\Sigma^-$ nature and would participate in the predissociation of the vibrational levels of the $A^2\Sigma^+$ state.

In the second part of this study, we will mainly focus on the X and A states of both ions. These states were reinvestigated using internally contracted multireference configuration interaction (MRCI) computations.^{16,17} During these calculations, the active space was reduced to $(10\sigma, 4\pi)$ MOs, with the $(4\sigma, 1\pi)$ MOs frozen and the $(6\sigma, 2\pi)$ MOs restricted to be doubly occupied. Finally, the Davidson correction¹⁸ which approximates the contribution of quadruple excitations, has been included (MRCI+Q). Figures 2 and 3 present the resulting electronic states of

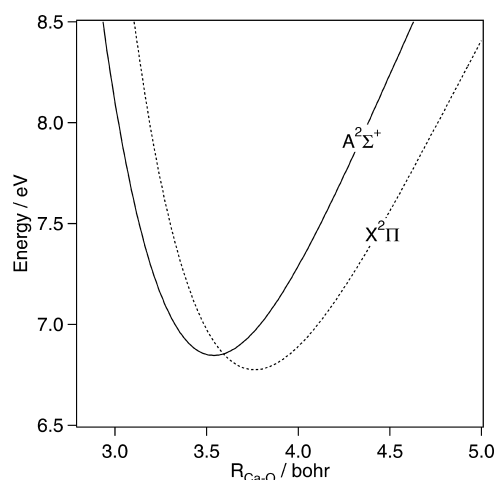


Figure 2. MRCI+Q potential energy curves of the $X^2\Pi$ and $A^2\Sigma^+$ states of CaO^+ . The reference energy corresponds to the minimum of the electronic ground state of the neutral molecule.

CaO^+ and CaO^- , respectively. For CaO^- , we also reported the curves of the low-lying electronic states of CaO computed at the same level of theory.¹ For this anion, comparison between the CASSCF (Figure 1) and MRCI+Q (Figure 3) curves shows similarities with the neutral molecule:¹ at the CASSCF level, the X and A states appear to be quasi-degenerated, whereas these states are well-separated at the MRCI+Q level. Such correlation effects were also found for the neutral molecule¹ and were explained by the multiconfigurational character of the X state. For the CaO^+ cation, both the CASSCF and the MRCI+Q curves reveal that the X and A states remain close in energy. The removal of one electron compared with the neutral molecule clearly simplified the description of the electronic wave function, and the multiconfigurational character is strongly reduced. These

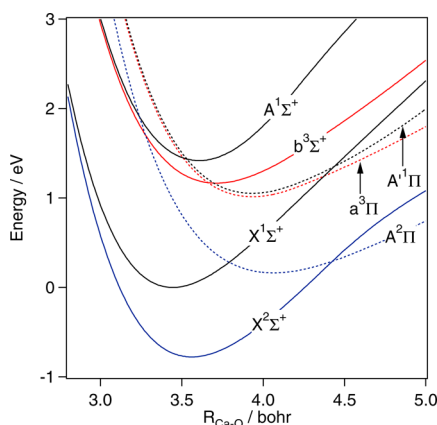


Figure 3. MRCI+Q potential energy curves of the $X^2\Sigma^+$ and $A^2\Pi$ states of CaO^- and of the low-lying electronic states $X^1\Sigma^+$, $A^1\Sigma^+$, $b^3\Sigma^+$, $a^3\Pi$, and $A^1\Pi$ of neutral CaO . The reference energy corresponds to the minimum of the electronic ground state of the neutral molecule.

states are, however, coupled together by SO couplings, and interactions are expected even between the lowest vibrational levels of these states. A rigorous treatment of these couplings will be presented in the next section.

Table 2 presents the main spectroscopic constants of the ionic electronic states computed with the MRCI+Q potential energies

Table 2. Main Spectroscopic Parameters of the Low-Lying $^2\Sigma^+$ and $^2\Pi$ States of CaO^+ and CaO^-

CaO^-	$R_e/\text{\AA}$	ω_e/cm^{-1}	$\omega_e x_e/\text{cm}^{-1}$	B_e/cm^{-1}	D_e/eV	T_0/eV
$X^2\Sigma^+$	1.884	681.8	3.3	0.4156	3.60 ^a	0.0
calc. ⁴	1.893	671			3.54	0.0
$A^2\Pi$	2.153	464.8	3.4	0.3185	2.66	0.93
calc. ⁴	2.151	467			2.79	0.72
CaO^+	$R_e/\text{\AA}$	ω_e/cm^{-1}	$\omega_e x_e/\text{cm}^{-1}$	B_e/cm^{-1}	D_e/eV	T_0/eV
$X^2\Pi$	1.990	639.3	2.8	0.3725	3.43 ^b	0.0
calc. ⁴	1.992	638			3.33	0.0
calc. ³	1.996	669			3.29	0.0
$A^2\Sigma^+$	1.873	703.2	3.0	0.4205	5.05 ^c	0.07
calc. ⁴	1.875	704			4.94	0.08
calc. ³	1.885	728			3.29	0.09

^aFrom the calculation of $\text{Ca}(^1S) + \text{O}^-(^2P)$. ^bFrom the calculation of $\text{Ca}^+(^2S) + \text{O}(^3P)$. ^cFrom the calculation of $\text{Ca}^+(^2S) + \text{O}(^3P) +$ the average excitation $\text{Ca}^+(^2S) \rightarrow \text{Ca}^+(^2D)$ from ref 23.

using the Cooley algorithm.¹⁹ The calculated constants are the equilibrium bond length (R_e in $\text{\AA}$), the harmonic and anharmonic vibrational wavenumbers (ω_e and $\omega_e x_e$ in cm^{-1}), the rotational constant (B_e in cm^{-1}), and the adiabatic excitation energy (T_0 in eV, including the zero-point energy correction). For the CaO^- anion, the equilibrium geometries and harmonic wavenumbers are found to be consistent with the value predicted by Harrison et al.⁴ using the coupled cluster theory, with a difference of $<0.01 \text{ \AA}$ for the equilibrium geometry of the $X^2\Sigma^+$ state and a difference of $0.002 \text{ \AA}$ for the $A^2\Pi$ state. For ω_e , the difference is $<10 \text{ cm}^{-1}$ for the X state and $<3 \text{ cm}^{-1}$ for the $A^2\Sigma^+$ state. This difference for the X state could be explained by the slight multiconfigurational character (with a dominant electron configuration weight of 0.92 at the MRCI level) of this electronic ground state in the molecular region, as previously found for the electronic ground state of the neutral CaO :¹ a similar difference was found between the harmonic frequency computed with our MRCI+Q potential

energy curves and the one deduced by Harrison et al.⁴ from their RCCSD(T) curve. According to the agreement between the experimental determination and our MRCI+Q result in ref 1 for the neutral electronic ground state, we are confident concerning the anion value. The adiabatic excitation energy of the $A^2\Pi$ state of CaO^- is deduced at 0.93 eV, $\sim 0.21 \text{ eV}$ larger than the value computed by Harrison et al.⁴ Here again this difference could be explained by the multiconfigurational character of the X state and a deeper potential well computed using our MRCI+Q approach than the one obtained in coupled cluster calculations. It can be noticed that using our level of theory the $A^2\Pi$ state is found to be stable but is reached for energies higher than the EA of CaO , which we calculated at 0.79 eV using the MRCI+Q method. For comparison, Harrison et al.⁴ reported an EA of 0.865 eV.

Turning now to the discussion on the CaO^+ cation, the spectroscopic constants deduced from our MRCI+Q curves for the X and A states (Table 2) are close to those obtained by Harrison et al.,⁴ confirming the single-configuration character of these two states. The equilibrium geometries and electronic transition energies obtained by Partridge et al.³ with single- and double-configuration interaction calculations are also consistent, but a difference of $\sim 30 \text{ cm}^{-1}$ is noticed for the harmonic wavenumbers of both states. Using our level of calculations, the adiabatic IE of CaO , that is, $\text{CaO}(X^1\Sigma^+) \rightarrow e^- + \text{CaO}^+(X^2\Pi)$, is computed at 6.76 eV, in remarkable agreement with the 6.767 eV value computed by Harrison et al.⁴ and consistent with the value ranging between 6.5 and 6.7 eV deduced from mass spectrometric studies.^{5–7} The adiabatic excitation energy between the X and A states of CaO^+ is obtained at 0.07 eV, in agreement with the previous theoretical works.^{3,4}

The analysis of the MOs reveals that the 8σ and 3π MOs are mainly associated with the $2p$ atomic orbital (AO) of O, and the 9σ MO with the $4s$ AO of Ca. The main electron configuration of the ionic and neutral electronic states at $R = 4 \text{ bohr}$ are given by: for CaO^+ , $X^2\Pi$ ($\dots 8\sigma^2 3\pi^3$) and $A^2\Sigma^+$ ($\dots 8\sigma^1 3\pi^4$), associated with a Ca^{2+}O^- configuration; for CaO , $X^1\Sigma^+$ ($\dots 8\sigma^2 3\pi^4$) and ($\dots 8\sigma^1 3\pi^4 9\sigma^1$), associated with the $\text{Ca}^{2+}\text{O}^{2-}$ and Ca^+O^- configurations, respectively, $b^3\Sigma^+$ ($\dots 8\sigma^1 3\pi^4 9\sigma^1$), $A^1\Pi$ and $a^3\Pi$ ($\dots 8\sigma^2 3\pi^3 9\sigma^1$), associated with a Ca^+O^- configuration; and for CaO^- , $X^2\Sigma^+$ ($\dots 8\sigma^2 3\pi^4 9\sigma^1$) and $A^2\Pi$ ($\dots 8\sigma^2 3\pi^3 9\sigma^2$), associated with Ca^+O^{2-} and CaO^- configurations, respectively. The 9σ electron is then removed to ionize the neutral CaO and the additional electron of CaO^- is associated either with the 3π or 9σ orbitals. This analysis can help to explain the values of the ionization energy, IE = 6.76 eV, and of the electron affinity, EA = 0.79 eV, of CaO . The IE of CaO is close to the IE of Ca (IE(Ca) = 6.11 eV²⁰) and the EA of CaO is almost the average of EAs of oxygen and calcium (EA(O) = 1.46 eV,²¹ EA(Ca) = 0.043 eV²²).

3. VIBRONIC STATES

The SO interactions between the electronic components of the ions imply the use of the resulting quantum number Ω , where

$$\Omega = \Lambda + \Sigma \quad (1)$$

with Λ and Σ , the quantum numbers associated with the projection on the z axis of the electronic and spin angular momenta, respectively.

The total Hamiltonian, $\hat{H} = \hat{T}_N + \hat{H}_e + \hat{H}_{\text{SO}}$ is then represented on the spin-complex electronic eigenfunctions $\phi_i^{\Lambda, S, \Sigma}$ such that the total eigenfunction can be written as

$$\Psi(\mathbf{r}, R) = \sum_i C_{iv} \phi_i^{\Lambda, S, \Sigma}(\mathbf{r}, R) \frac{\chi_i^v(R)}{R} \quad (2)$$

where $\mathbf{r}$ is the electron coordinates, R is the nuclear separation, and $\chi_i^v(R)$ is the solution of the vibrational Schrödinger equation for the electronic state i associated with the potential energy function V_i

$$\hat{H}_v = -\frac{1}{2\mu} \frac{d^2}{dR^2} + V_i \quad (3)$$

μ is the reduced mass of the diatomic molecule. In both ions, the rotational part of the nuclear kinetic energy operator $\hat{T}_N$ was not taken into account because our previous work on CaO showed that the effect of rotation can be neglected with respect to the SO effect.² This is also confirmed by the low values of the rotational constants, B_v , displayed in Table 2.

The SO coupling terms within the $X^2\Pi$ and $A^2\Sigma^+$ electronic states of CaO^+ were calculated at the MRCI level using the Breit–Pauli Hamiltonian, $\hat{H}_{\text{SO}} = \hat{H}^{\text{BP}}$, implemented in MOLPRO.^{15,24} The same active space and basis sets as for the potentials have been used. The nonzero expectation values of the Breit–Pauli Hamiltonian represented on the basis of the spin-real electronic eigenfunctions of the electronic Hamiltonian can be obtained from SO_z and SO_x

$$\begin{aligned} \text{SO}_z &= i \langle {}^2\Pi_x, \Sigma = 1/2 | \hat{H}_{\text{SO}}^{\text{BP}} | {}^2\Pi_y, \Sigma = 1/2 \rangle \\ \text{SO}_x &= \langle {}^2\Sigma^+, \Sigma = 1/2 | \hat{H}_{\text{SO}}^{\text{BP}} | {}^2\Pi_x, \Sigma = -1/2 \rangle \end{aligned} \quad (4)$$

where the z direction is taken as the molecular axis. The variation of the resulting terms with R is presented in Figure 4.

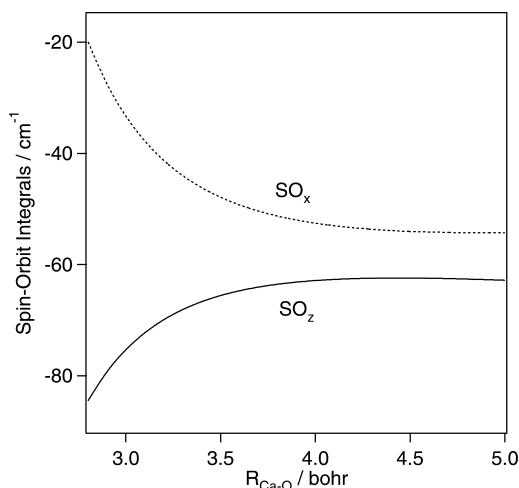


Figure 4. SO coupling terms resulting from the evaluation of the expectation values of the Breit–Pauli Hamiltonian.

The representation of operator $\hat{H}$ on the basis of the spin-complex electronic eigenfunctions $\phi_i^{\Lambda, S, \Sigma}$ for CaO^+ results in coupling matrices, that is, for $\Omega = \pm 1/2$

$$H_{|\Omega|=1/2} = -\frac{1}{2\mu} \frac{d^2}{dR^2} \mathbf{I} + \begin{pmatrix} V_{2\Pi} - \text{SO}_z & \text{SO}_x \\ \text{SO}_x & V_{2\Sigma^+} \end{pmatrix} \quad (5)$$

and for $\Omega = \pm 3/2$ (single term)

$$H_{|\Omega|=3/2} = -\frac{1}{2\mu} \frac{d^2}{dR^2} + (V_{2\Pi} + \text{SO}_z) \quad (6)$$

as demonstrated in ref 25.

For CaO^- , the SO coupling is not taken into account because the difference between the $X^2\Sigma^+$ and $A^2\Pi$ energy minima is larger than the EA. It can be mentioned, as discussed in Section 2, that the $A^2\Pi$ anionic state is crossing the $X^1\Sigma^+$ state for $R = 3.8$ bohr approximately, and the autoionization process that should occur is not treated in the present work. The matrix used for the determination of the vibrational energies of the $X^2\Sigma^+$ state of CaO^- is then the single term

$$H_{|\Omega|=1/2} = -\frac{1}{2\mu} \frac{d^2}{dR^2} + V_{2\Sigma^+} \quad (7)$$

For both ions, the vibronic energies have been determined using a Lanczos diagonalization.²⁶

In the one-electron approximation²⁷

$$\langle {}^2\Pi_{3/2}(\text{CaO}^+) | \hat{H}_{\text{SO}} | {}^2\Pi_{3/2}(\text{CaO}^+) \rangle = \text{SO}_z \approx -\frac{a_\pi}{2} \quad (8)$$

with

$$a_\pi = \langle 3\pi | \hat{h}_{\text{SO}} | 3\pi \rangle \approx \langle 2p | \hat{h}_{\text{SO}} | 2p \rangle \quad (9)$$

and $\hat{h}_{\text{SO}}$ is the one-electron SO operator. In CaO, we have previously shown that the internal SO coupling term of the $a^3\Pi$ state, noted SO_5 in ref 2, can also be expressed in the one-electron approximation such that

$$\text{SO}_5 = \langle {}^3\Pi_{\Omega=2}(\text{CaO}) | \hat{H}_{\text{SO}} | {}^3\Pi_{\Omega=2}(\text{CaO}) \rangle \approx -\frac{a_\pi}{2} \quad (10)$$

At $R = 5.0$ bohr, $\text{SO}_z = -62.8 \text{ cm}^{-1}$ and $\text{SO}_5 = -61.2 \text{ cm}^{-1}$.² For the ${}^2\Pi$ state of CaO^- , we also calculated that $\langle {}^2\Pi_{3/2}(\text{CaO}^-) | \hat{H}_{\text{SO}} | {}^2\Pi_{3/2}(\text{CaO}^-) \rangle = -62.8 \text{ cm}^{-1}$ at the equilibrium geometry of the state, that is, $R = 4.068$ bohr. These three values of SO terms show that the SO coupling in the neutral and ionic Π states is mainly due to the three electrons in the 3π orbital for $R > 4$ bohr (see configurations in the previous section) and that the one-electron approximation, based on the fact that the 3π MO can be associated with the $2p$ oxygen orbital for larger values of R , is valid. A limitation of this one-electron approximation is notable for the transition term SO_x , which should be associated with $-a_\pi/2\sqrt{2} = -44.4 \text{ cm}^{-1}$ but is calculated to be -54.3 cm^{-1} at 5 bohr.

The resulting vibronic energies of the $X^2\Pi$ and $A^2\Sigma^+$ electronic states of the CaO^+ are given in Table 3, with or without considering the SO couplings. IE equals $54\,817.4 \text{ cm}^{-1} = 6.7965 \text{ eV}$ in agreement with the adiabatic electronic value and is reduced to $54\,753.7 \text{ cm}^{-1} = 6.7886 \text{ eV}$ if the SO splitting of the ${}^2\Pi$ state is taken into account. The final splitting between the $|\Omega| = 3/2$ and $1/2$ $v = 0$ levels is 125.2 cm^{-1} . The harmonic and anharmonic wavenumbers stay unchanged compared with the values of Table 2, even if the internal SO_z coupling is added, but important changes appear if SO_x is included in matrix (5). The first levels of the $X^2\Pi$ and $A^2\Sigma^+$ states are close in energy and interact due the SO couplings. In Figure 5, the energy differences between SO coupled and uncoupled vibronic levels of the $X^2\Pi$ and $A^2\Sigma^+$ states of CaO^+ are plotted according to the uncoupled energy levels for $|\Omega| = 1/2$. The largest deviations of $\sim 6 \text{ cm}^{-1}$ are found for the $v^+ = 1, 12$ of state $X^2\Pi$ and for $v^+ = 0, 10$ of state $A^2\Sigma^+$.

Table 3. Vibrational Levels (in cm^{-1}) of the Two Lowest States of CaO^+ , $X^2\Pi$, and $A^2\Sigma^+$ ^a

ν^+	$X^2\Pi$	$X^2\Pi_{ \Omega =3/2}$	$X^2\Pi_{ \Omega =1/2}$	$X^2\Pi_{ \Omega =1/2}$ coupled	$A^2\Sigma^+$	$A^2\Sigma^+_{ \Omega =1/2}$ coupled
0	54817.4	54753.7	54881.1	54878.9	55415.3	55408.6
1	55451.2	55387.5	55514.9	55520.5	56112.6	56111.7
2	56079.5	56015.8	56143.3	56143.9	56804.1	56807.0
3	56702.4	56638.7	56766.2	56763.6	57489.7	57492.5
4	57320.0	57256.2	57383.8	57381.1	58169.6	58170.9
5	57932.2	57868.4	57996.0	57994.5	58843.7	58843.9
6	58539.1	58475.3	58603.0	58602.5	59512.1	59512.0
7	59140.8	59077.0	59204.7	59204.7	60174.9	60175.0
8	59737.4	59673.5	59801.3	59801.3	60832.1	60832.6
9	60328.7	60264.8	60392.7	60392.4	61483.9	61484.2
10	60914.7	60850.8	60978.7	60978.2	62130.1	62124.6
11	61495.3	61431.3	61559.3	61558.9	62770.9	62772.9
12	62070.4	62006.4	62134.4	62140.0	63406.4	63407.6
13	62640.1	62576.1	62704.1	62702.2	64036.5	64037.2
14	63204.5	63140.5	63268.6	63267.3	64661.3	64661.5
15	63763.9	63699.9	63828.0	63827.2		
16	64318.4	64254.4	64382.5	64382.1		
17	64868.2	64804.1	64932.3	64932.2		

^aEnergy reference is the $\nu = 0$ level of $X^1\Sigma^+$ of CaO . If Ω is not specified, then the levels are determined from the treatment of Hamiltonian (3). $|\Omega| = 1/2$ values are obtained from diagonal part of eq 5 and $|\Omega| = 1/2$ “coupled” values from the fully coupled eq 5. $|\Omega| = 3/2$ values are obtained from the treatment of Hamiltonian (6).

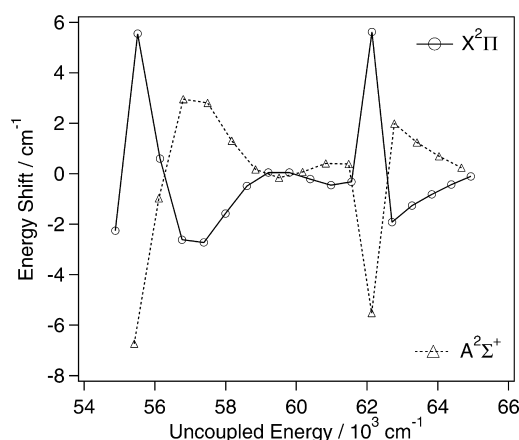


Figure 5. Energy differences between SO coupled and uncoupled vibronic levels of the $X^2\Pi$ and $A^2\Sigma^+$ states of CaO^+ as a function of the uncoupled energy levels for $|\Omega| = 1/2$. The full and dashed lines correspond to the $^2\Pi$ and $^2\Sigma^+$ level shifts, respectively.

For CaO^- , Table 4 gives the vibrational levels of the $X^2\Sigma^+$ state as well as those of the $X^1\Sigma^+$ state of CaO relative to the $\nu^- = 0$ level of CaO^- . The EA equals $6390 \text{ cm}^{-1} = 0.7923 \text{ eV}$, in agreement with the adiabatic electronic value. No vibrational level is given for the excited state of CaO^- because the $A^2\Pi$ anionic state is subject to an electron detachment for R shorter than 3.8 bohr. Additionally, it can be noticed that there are differences of a few wavenumbers between the vibrational levels of neutral CaO given in Table 4 compared with our previous work.² These differences are due to copying errors of preliminary results in ref 2. The present values in Table 4 have been carefully checked and are correct.

4. PHOTOELECTRON SPECTRA OF CaO^- AND CaO

We performed wavepacket propagations^{28,29} to compute the photoelectron spectrum $\text{CaO} (X^1\Sigma^+) \rightarrow e^- + \text{CaO}^+ (X^2\Pi, A^2\Sigma^+)$, from a Fourier transform of the autocorrelation function, $\langle \Psi(0) | \Psi(t) \rangle$. The propagations were done for $|\Omega| = 1/2$ using

Table 4. Vibrational Levels (in cm^{-1}) of the Electronic Ground States $X^1\Sigma^+$ of CaO and $X^2\Sigma^+$ of CaO^- Using Hamiltonian (3)^a

ν or ν^-	$X^2\Sigma^+ \text{ CaO}^-$	$X^1\Sigma^+ \text{ CaO}$
0	0.0	6390.0
1	675.2	7117.1
2	1343.5	7836.1
3	2005.0	8544.1
4	2659.6	9240.6
5	3307.3	9926.2
6	3948.3	10601.8
7	4582.5	11267.6
8	5210.0	11924.1
9	5830.8	12571.2

^aThe energy reference is the $\nu^- = 0$ level of $X^2\Sigma^+$ of CaO^- .

the Hamiltonian operator of eq 5, respectively, with durations up to 4 ps using a time step of $\delta t = 0.2 \text{ fs}$ and 128 grid points with $R \in [2.8; 5] \text{ bohr}$. The detached electron is considered to instantaneously leave the molecule and is not described. The transition moment between the neutral and the ionic electronic states was taken to unity and kept constant. The initial wavepacket, $|\Psi(0)\rangle$, is the combination of the first vibrational levels of the $X^1\Sigma^+$ state of CaO , populated according to the Boltzmann statistics for a temperature of 300 K.

Two initial conditions were considered, with $|\Psi(0)\rangle$ being promoted entirely either on the Σ or on the Π state. The binding electron intensities resulting from Fourier transform are just proportional to the overlap between the initial wave function and the excited bound vibronic states. It must be noticed that all spectra obtained at 300 K are very similar to those calculated at 0 K because the Boltzmann distribution gives a population larger than 95% of the ground vibronic state at 300 K.

In Figure 6, the two photoelectron spectra so obtained for CaO ($|\Delta\Omega| = 1/2$) are displayed. Because the SO coupling is small, one is mostly related to $\text{CaO} (X^1\Sigma^+) \rightarrow e^- + \text{CaO}^+ (X^2\Pi_{1/2})$ and the other to $\text{CaO} (X^1\Sigma^+) \rightarrow e^- + \text{CaO}^+ (A^2\Sigma^+_{1/2})$.

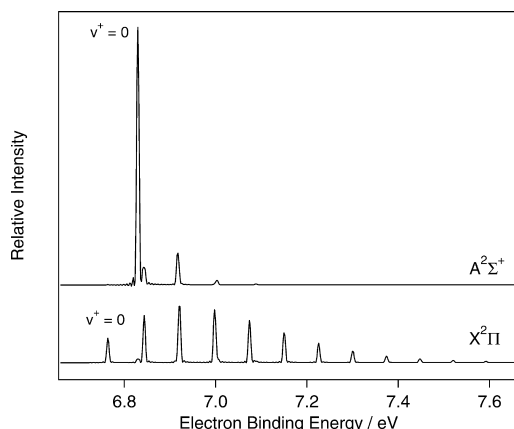


Figure 6. Simulated $\text{CaO}(\text{X}^1\Sigma^+) \rightarrow e^- + \text{CaO}^+$ photoelectron spectra at $T = 300$ K and for $|\Delta\Omega| = 1/2$. The energy reference is the $\text{CaO}(\text{X}^1\Sigma^+, \nu = 0)$ vibrational level.

The harmonic wavenumbers of $\text{X}^1\Sigma^+(\text{CaO})$, $\text{X}^2\Pi$, and $\text{A}^2\Sigma^+(\text{CaO}^+)$ are 732.7, 639.3, and 703.2 cm^{-1} , and the equilibrium geometries are $R_e = 1.819$, 1.990, and 1.873 Å, respectively. So, the Franck–Condon region of the electronic ground state of CaO has a larger overlap with the vibrational levels of the cationic $\text{A}^2\Sigma^+$ state. The consequences are that the photoelectron spectrum leading to the $\text{A}(\text{CaO}^+)$ state exhibits few intense peaks with a maximum for $\nu^+ = 0$, whereas the photoelectron spectrum leading to the $\text{X}(\text{CaO}^+)$ state presents a longer less intense progression with a maximum for $\nu^+ = 2$. The coupling between both X and A electronic states has a signature in both spectra as the presence of a splitting of the peaks. The $|\Omega| = 3/2$ photoelectron spectrum has not been simulated because it must be very similar to the $\text{CaO}(\text{X}^1\Sigma_0^+) \rightarrow e^- + \text{CaO}^+(\text{X}^2\Pi_{1/2})$ one, without the splittings (Hamiltonian operator of eq 6).

For the CaO^- photoelectron spectrum, the situation is more complex. The low-lying electronic states of CaO have been described in details in ref 1 and in Figure 3. The spin–orbit couplings imply that Ω can have the values ± 2 , ± 1 , 0^+ , and 0^- , such that four coupling Hamiltonian matrices must be used for the propagation. These matrices can be derived from ref 2 and are:

$$\begin{aligned}
 H_{|\Omega|=2} &= -\frac{1}{2\mu} \frac{d^2}{dR^2} + (V_{3\Pi} + SO_5) \\
 H_{|\Omega|=1} &= -\frac{1}{2\mu} \frac{d^2}{dR^2} \mathbf{I} + \begin{pmatrix} V_{1\Pi} & -\sqrt{2}SO_2 & -SO_3 \\ -\sqrt{2}SO_2 & V_{3\Sigma^+} & -\sqrt{2}SO_4 \\ -SO_3 & -\sqrt{2}SO_4 & V_{3\Pi} \end{pmatrix} \\
 H_{\Omega=0^+} &= -\frac{1}{2\mu} \frac{d^2}{dR^2} \mathbf{I} + \begin{pmatrix} V_{3\Pi} - SO_5 & -2SO_{1A} & -2SO_1 \\ -2SO_{1A} & V_{A^1\Sigma^+} & 0 \\ -2SO_1 & 0 & V_{X^1\Sigma^+} \end{pmatrix} \\
 H_{\Omega=0^-} &= -\frac{1}{2\mu} \frac{d^2}{dR^2} \mathbf{I} + \begin{pmatrix} V_{3\Sigma^+} & -2SO_4 \\ -2SO_4 & V_{3\Pi} - SO_5 \end{pmatrix}
 \end{aligned} \quad (11)$$

with the SO_i coupling terms defined in ref 2. As in ref 2, the excited states of CaO have been shifted with respect to the electronic ground state such that the ground vibronic level of

each state has the same energy as the experimental measurement. However, it must be noticed that the shifts of ref 2 must be replaced by the correct values: 307.3 cm^{-1} for the $a^3\Pi$, 300.9 cm^{-1} for the $A^1\Pi$, 245.6 cm^{-1} for the $b^3\Sigma^+$, and 209.3 cm^{-1} for the $A^1\Sigma^+$.

Using the same methodology as previously described, we simulated the photoelectron spectra of CaO^- (reported in Figure 7). Four spectra have been superimposed. A zoom related to the

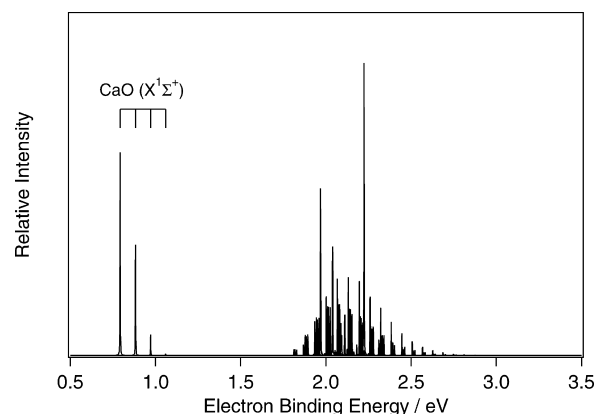


Figure 7. $\text{CaO}^-(\text{X}^2\Sigma^+) \rightarrow e^- + \text{CaO}$ photoelectron spectrum at $T = 300$ K. The energy reference is the $\text{CaO}^-(\text{X}^2\Sigma^+, \nu^- = 0)$ vibrational level.

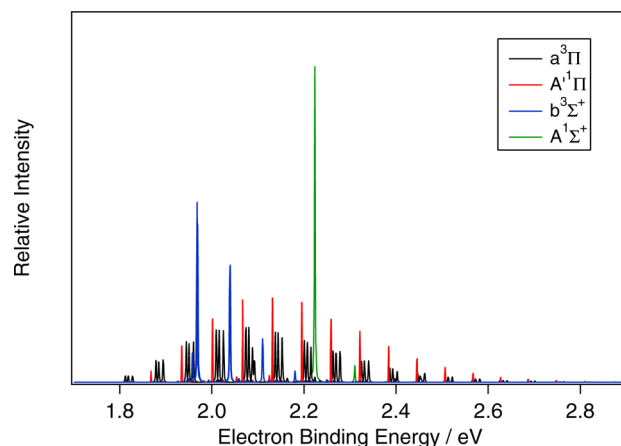


Figure 8. Zoom of the $\text{CaO}^-(\text{X}^2\Sigma^+) \rightarrow e^- + \text{CaO}$ photoelectron spectrum at $T = 300$ K around 1.8 and 2.8 eV. The energy reference is the $\text{CaO}^-(\text{X}^2\Sigma^+, \nu^- = 0)$ vibrational level.

excited states of CaO is also given in Figure 8. The remarkable points are: (i) very intense and short progressions for electron detachment toward the triplet and singlet Σ^+ states of CaO, (ii) low intense and long progressions for electron detachment toward the triplet and singlet Π states of CaO, and (iii) the interacting vibronic levels via SO coupling found in ref 2 are also observed in the spectra by the presence of multiplet peaks. The strongest couplings appear close to the $\nu = 0$ peak of $b^3\Sigma^+$ at 1.96 eV and close to the $\nu = 0$ peak of $A^1\Sigma^+$ at 2.22 eV. The analysis of the corresponding SO couplings has been detailed in ref 2. The main SO effect can be seen in the internal splitting of the $a^3\Pi$ state. As in the CaO photoelectron spectrum, the explanations for points (i) and (ii) are related to the difference of equilibrium geometries and harmonic wavenumbers of the anionic $\text{X}^2\Sigma^+$ electronic ground state and of the neutral electronic states.^{1,2}

5. CONCLUSIONS

The first two electronic states of CaO^+ and CaO^- have been computed at the MRCI+Q level of theory, which was recently used for the accurate determination of the five lowest electronic states of CaO . The same active spaces have been used, and the present calculations lead to $\text{IE}(\text{CaO}) = 6.79$ eV and $\text{EA}(\text{CaO}) = 0.79$ eV. The first electronic states of CaO^+ are doublet and exhibit a spin–orbit coupling and splitting similar to the ones of the neutral states, for which the additional electron is located in the quasi-nonbonding 9σ orbital. The spin–orbit coupling can be observed in resulting photoelectron spectra of CaO^+ , where splittings of the interacting vibronic peaks are present. For the anion, the $X^2\Sigma^+$ electronic ground state is well-isolated, whereas the first excited $A^2\Pi$ state is located at almost the same energy as the CaO electronic ground state with an increase of more than 0.6 bohr in the equilibrium bond length. This state is then unstable with respect to the electron detachment, and the predissociation of the vibronic states should be treated with more sophisticated methods. A photoelectron spectrum of CaO^- up to 3.5 eV, involving the five lowest electronic states of CaO , has also been simulated in the aim of a complete predictive study. Indeed, the present simulated photoelectron spectra of CaO and CaO^- could be useful for further experimental studies that could be of the same type as those realized on MgO_{1-4}^- using photoelectron imaging spectroscopy.³⁰

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Notes

The authors declare no competing financial interest.

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